AUTONOMOUS DRIVING VEHICLES

CSU Sydney campus (202590)

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SUBJECT CODE: ITC105

SUBJECT NAME: Communication and Information management

Key concepts:

- History of Autonomous cars
  - Early Concepts and Prototypes
  - Different levels of vehicle autonomy (Level 0–5).
- Impact of self driving vehicles for elderly drivers.
  - Benefits and impacts on society
  - Driving decision making of autonomous vehicles.
- Cybersecurity of Autonomous vehicles
  - Threats to AV Communication Systems
  - Data Privacy and Personal Information Protection
- Human–Vehicle Interaction
  - Trust and user acceptance
  - Driver monitoring systems
  - Transition between manual and autonomous modes
- Legal and Ethical Challenges of Avs
  - Traffic rule interpretation
  - Legal responsibility in accidents
  - Ethical decision-making algorithms

Below are three recent academic relevant articles for topics above:

Lajunen, T., & Sullman, M. J. M. (2021). Attitudes Toward Four Levels of Self-Driving Technology Among Elderly Drivers. Frontiers in Psychology, 12, Article 682973. https://doi.org/10.3389/fpsyg.2021.682973

This article provides a overview about different SAE levels of vehicle autonomy. Includes explanations about different levels of self driving vehicles and how helpful it is for elderly drivers. It will be helpful to support the section about topic 1 and 2 above.

Bhuiyan, H., Governatori, G., Rakotonirainy, A., Wong, M. W., & Mahajan, A. (2023). Driving decision making of autonomous vehicle according to Queensland overtaking traffic rules.

This article will be relevant for my report . It shows how this technology makes decisions and interpretate road rules which it is one of the major challenges for Autonomous vehicles.

Westbrook, T. (2025). Cyber Security Of Autonomous Vehicles: The Implications For City Planning. Journal of Strategic Security, 18(1), 77–88.

This article highlights the weakest cybersecurity risks for self driving vehicles, including some of the more knowest as spoofing, denial of service and electronic interferences. Emphasizing the need of more safety regulations and software updates.

Below are three recent grey literature articles:

Mobileye. (2023, February 27). A brief history of autonomous vehicles – from Renaissance to Reality. Mobileye Blog. https://www.mobileye.com/blog/history-autonomous-vehicles-renaissance-to-reality/

This grey literature article found on google explains easily how it has been the developments of autonomous cars across the history. It will help me to support my next work.

TNO. (2024, October 7). Safe co-drivership with AVs requires advanced Human-Vehicle Interaction and Driver Monitoring Systems. TNO Insights. https://www.tno.nl/en/newsroom/insights/2024/10/safe-co-drivership-avs/

I will use this source to support the section "*Driver monitoring systems*". The author explains the new challenges technology are facing and how drivers are moving from being drivers to supervisors.

Haghani, M. (2024, January 29). What moral principles should self-driving car algorithms follow in situations where harm is unavoidable? ABC Religion & Ethics. https://www.abc.net.au/religion/what-moral-principles-should-guide-self-driving-ai-cars/103403186/

I will use this resource to give another perspective about how autonomous cars make life and death decisions. This source will support writing my report about on ethical challenges and public perception of autonomous vehicles decision making.

Reflection:

Through this assessment, I got a much better understanding of how to conduct an academic research and organise information properly. At the beginning, I found it challenging to address articles that matched key concepts, but making use of keywords terms and using filters such as publication year, I learned how quickly I could identify relevant sources. This helped me understand how important it is to be precise when navigating academic databases like CSU Library and Google Scholar. While I was writing this assessment I realized how important academic literature and grey literature are, each contributes differently to develop a well-supported report.

Researching about AVs also deepened my understanding of the topic itself. This concept has many main topics and subtopics that gave a new perspective about how multidisciplinary it is. Each source I selected helped me see a new dimension of the concept, which made it easier to organise my topics logically.

Overall, I learnt how to evaluate sources, get relevant information, and link it to specific sections of my report plan. Now, I feel more confident about writing assessment 3 as I have found a path with a much clearer direction.